

Sales Performance Diagnostic Report

Problems - Opportunities - Actions

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Sales Performance Data Analysis – Project Overview

Why?

Retail businesses generate large volumes of transactional data, but identifying what truly drives sales and **profitability** is not always straightforward. Understanding these patterns is key to improving **decision-making** and business performance.

What for?

The goal of this project is to extract **actionable insights** from the Superstore dataset. Specifically, it aims to identify profitable and unprofitable products, analyze performance across regions and customer segments, and evaluate the impact of discounts on profit.

How?

The analysis follows a structured data science workflow: data cleaning and preprocessing, exploratory data analysis (EDA), and visualization. This approach allows for uncovering trends, detecting anomalies, and **supporting data-driven business recommendations**.

Dataset Introduction

Data selection , shape, columns, topic
and business context

DATASET SELECTION

- Dataset selected : **Superstore Dataset**
- Data Context:

Faced with growing demand and fierce market competition, an analysis of this superstore's data is being sought to understand its current situation and how, through this analysis, it can improve its present and future. This data repository contains information on product sales, regions, categories, customer segments, and profit/loss for each order.

DATASET OVERVIEW

File:

- Type: *.csv (standard)*
- Size: **2.3MB**
- Rows: $\approx$ 10.000 rows
- Columns: **21 columns**

Description :

- The Superstore dataset is a retail transactional dataset containing around 10,000 orders, with detailed information on sales, profit, and customer activity.
- It includes 21 columns covering orders, customers, products, and geographic data, along with key metrics such as sales, quantity, discount, and profit.
- The data spans roughly from 2014 to 2017, enabling time-based analysis of business performance and trends.

Columns:

- Selected columns : **15**
 - **Numerical** : row_id , order_id , order_date , customer_id , customer_name , product_id , product_name , sales , quantity , discount , profit
 - **Categorical** : segment , region , category , sub-category

DATA CLEANING

First step - Load the dataset Superstore & import libraries

Second step - Info of data load

Third step - Review the Info and decide on the working dataframe

- 1 - Normalize column names
- 2 - Cutting working dataframe – 15 selected columns

Fourth step - Review empty , null and invalid values in all columns of work dataframe.

- **FOR ALL COLUMNS**
 - Deletion of records that have NULL
 - Deletion of records that have EMPTY
- **Column QUANTITY**
 - Deletion of records that have values <0
- **Column DISCOUNT**
 - Deletion of records that have values out of interval 0..1
- **Column ORDER DATE**
 - Deletion of records that don't have correct date format *mm/dd/yyyy* or *m/d/yyyy*

FINAL RESULT

- **A DataFrame clean an prepared to work**



Data Quality Checklist

- No missing values, no empty values and not NULL
- Selected columns (15) are complete across all 9,994 rows.
- No duplicate rows
- Row ID is unique per transaction. Multiple rows per Order ID are correct (multi-item orders).
- Order Date converted to datetime.
- Irrelevant columns removed:
 - Ship date, Ship mode, Country, City, Postal Code, Product Name dropped — not needed for analysis.
- Negative profit is valid
- Profit ranges from −\$6,600 to +\$8,400. Negative values reflect real business losses — not errors.

Profit Distribution

INSIGHT

The profit distribution reveals a deep structural problem: many orders generate losses. The red dashed line marks break-even (\$0). Everything to the left is a loss.

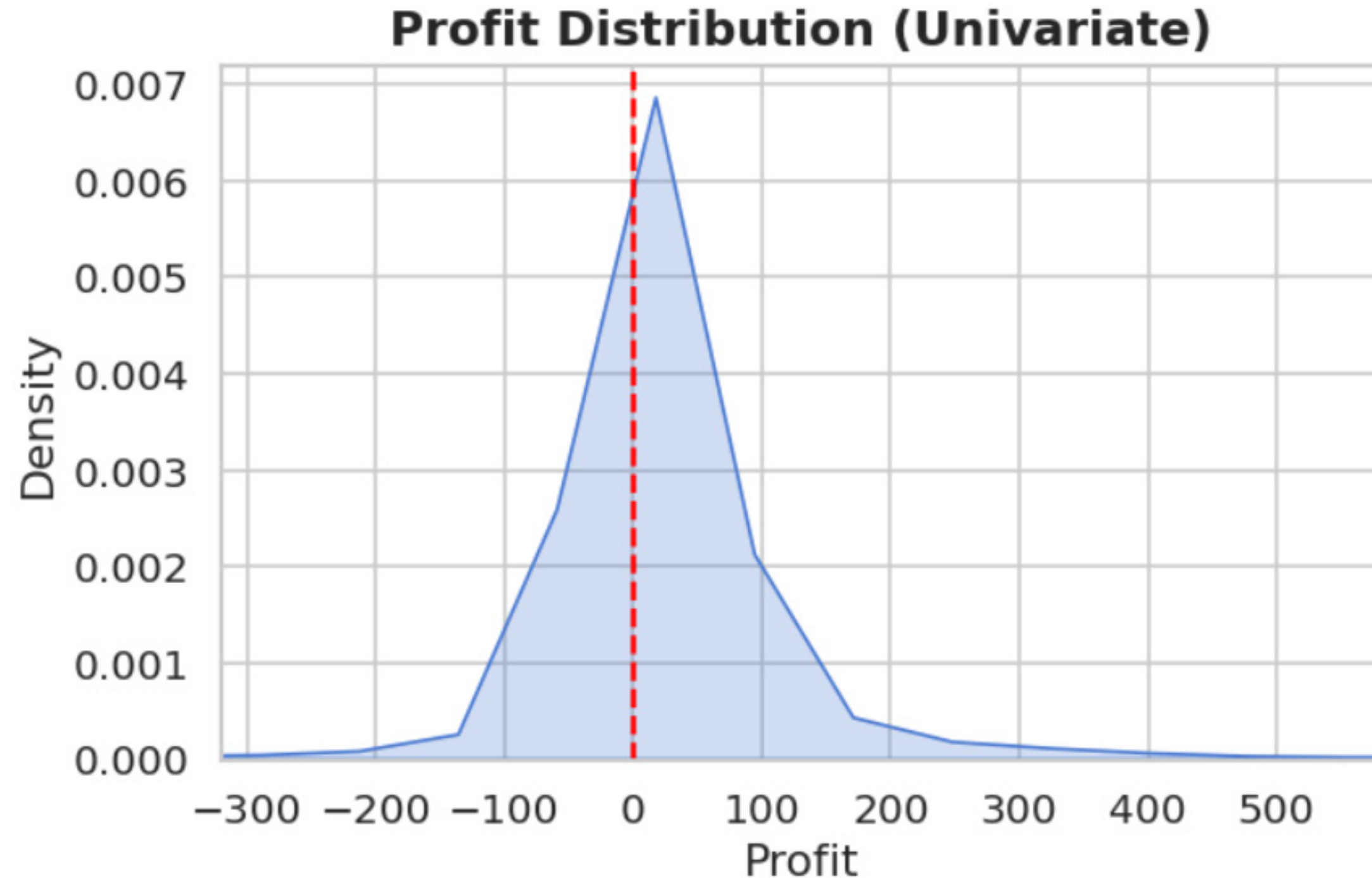
Distribution is centred near **\$0**

Long left tail → significant repeated losses

Minimum: **-\$6,600** in a single order

Maximum: **+\$8,400** in a single order

Many orders fall below the break-even line



Sales Distribution

INSIGHT

Sales per order are heavily right-skewed — the majority of orders are small, but a few very large orders pull the average up significantly.

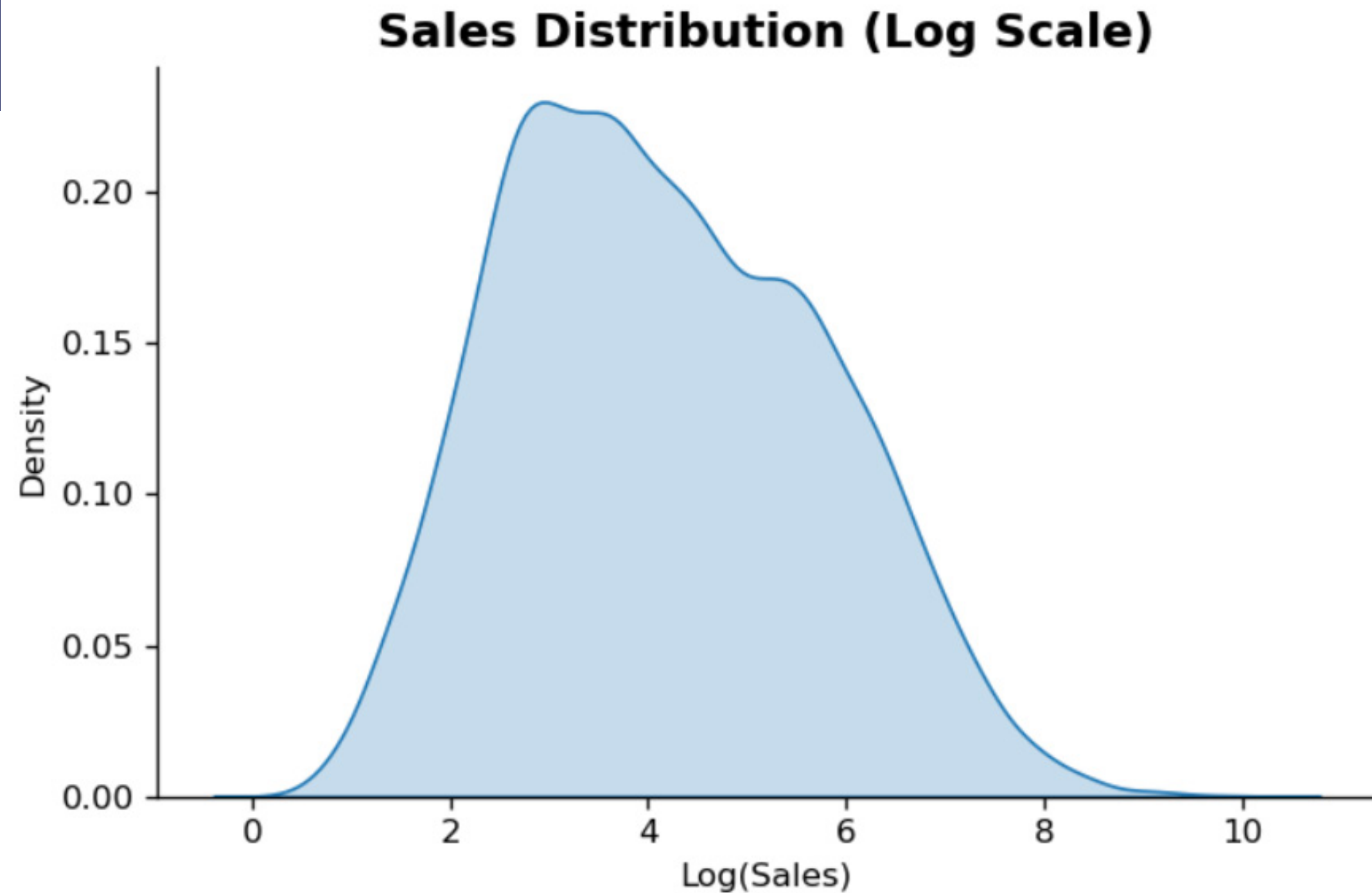
Most orders are between **\$10 and \$500**

Mean sales: **\$230** per order

A few outliers reach **+\$22,000**

High sales $\neq$ high profit — discount matters more

Skewed distribution $\rightarrow$ median better than mean



Discount Rate Distribution

INSIGHT

The discount rate is heavily concentrated at 0 — most orders have no discount at all. But a significant spike above 30% creates a dangerous tail.

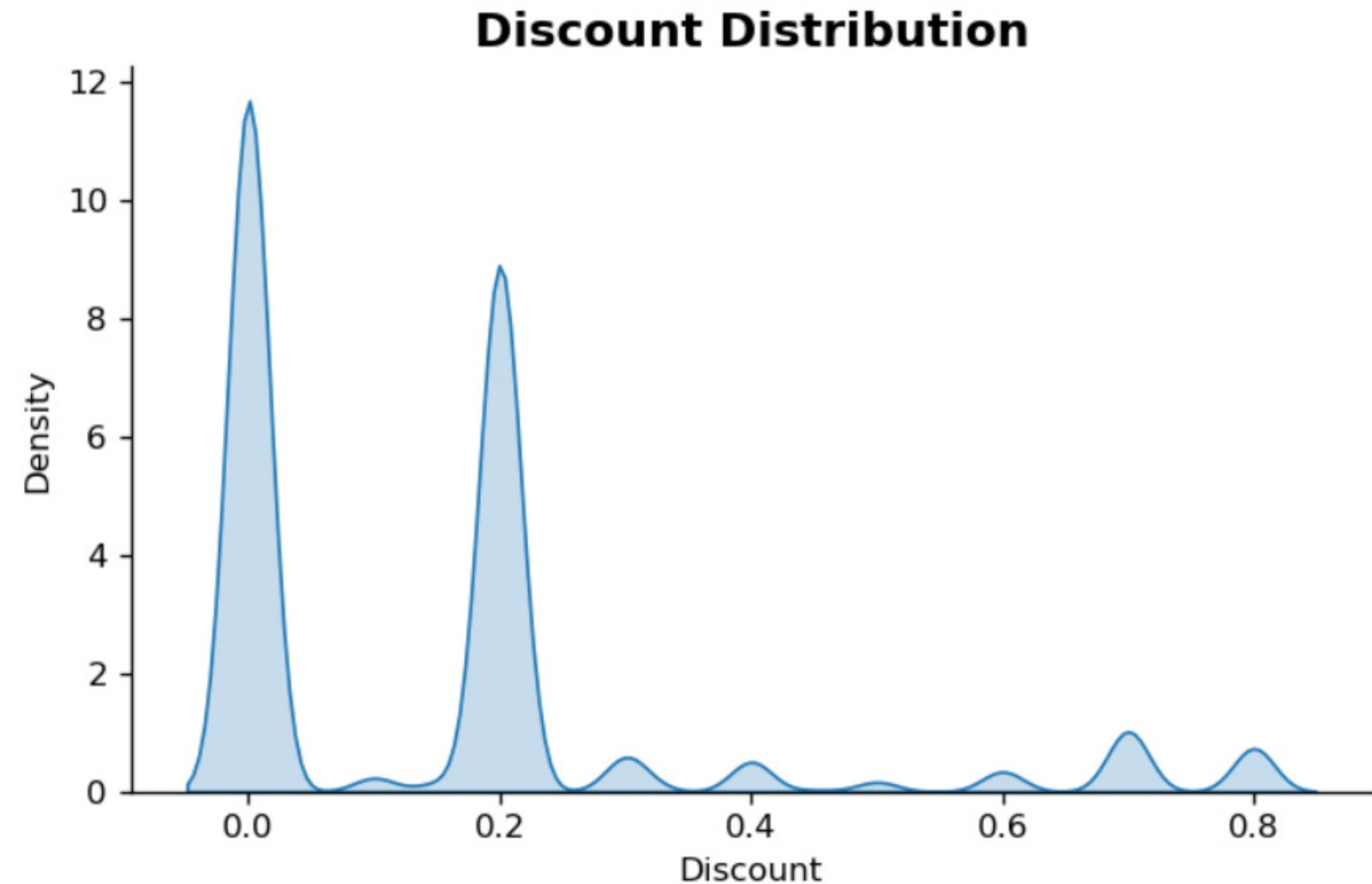
Most orders (**48%**) have **0% discount**

Many orders cluster at **20%**

A notable group receives **>30% discount**

The red line marks the 30% loss threshold

That tail of high discounts is where losses originate



Order Volume by Category

INSIGHT

Office Supplies dominates order count, but volume of orders does not equal profitability. Technology has fewer orders but much higher average order value.

Office Supplies: 6,026 orders (60%)

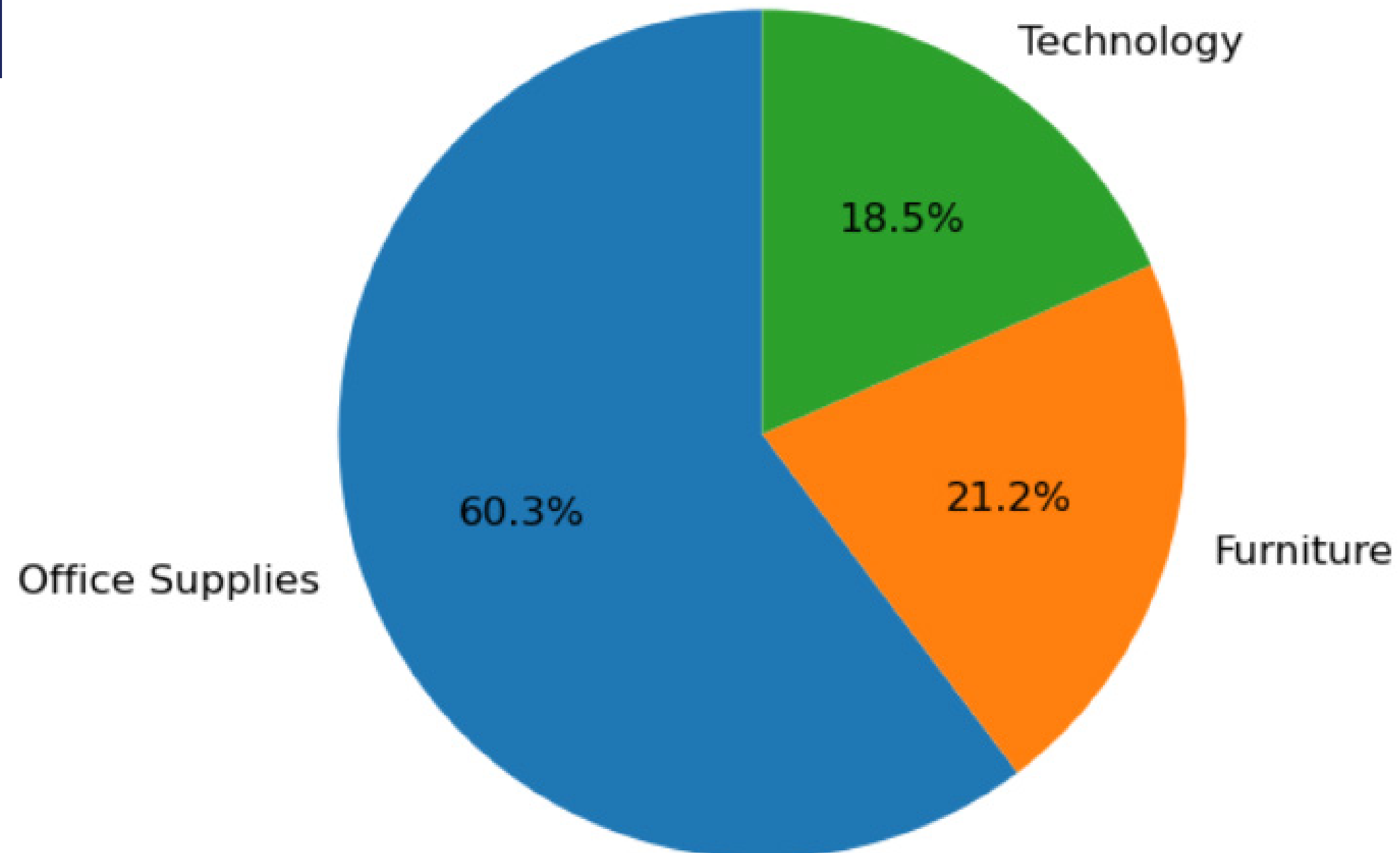
Furniture: 2,121 orders (21%)

Technology: 1,847 orders (19%)

Technology = high value, fewer orders

Volume and profitability tell different stories

Category Distribution

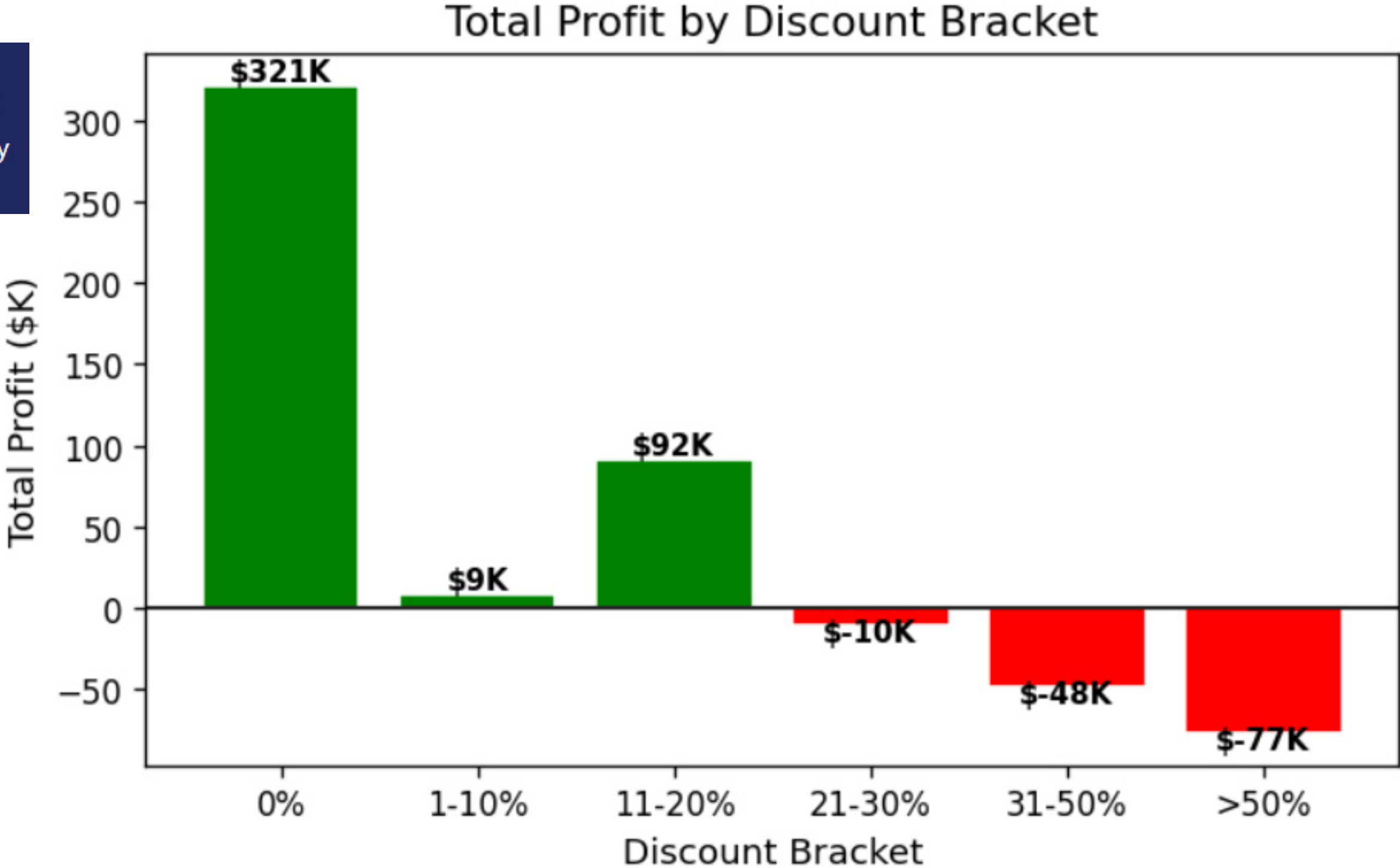


Discount vs. Profit

INSIGHT

Grouping orders into discount brackets makes the financial damage explicit and quantifiable. Green bars generate profit; red bars destroy it.

- 0% discount: +\$321K profit
- 1–20%: positive but declining
- 21–30%: –\$10K (loss territory begins)
- 31–50%: –\$48K
- >50% discount: –\$77K in losses
- No order above 30% discount is profitable in aggregate



Total Profit by Sub-Category

INSIGHT

Some sub-categories are highly profitable engines of the business. Others are structural money-losers regardless of region or customer. Red bars require immediate action.

Copiers: best performer at +\$56K

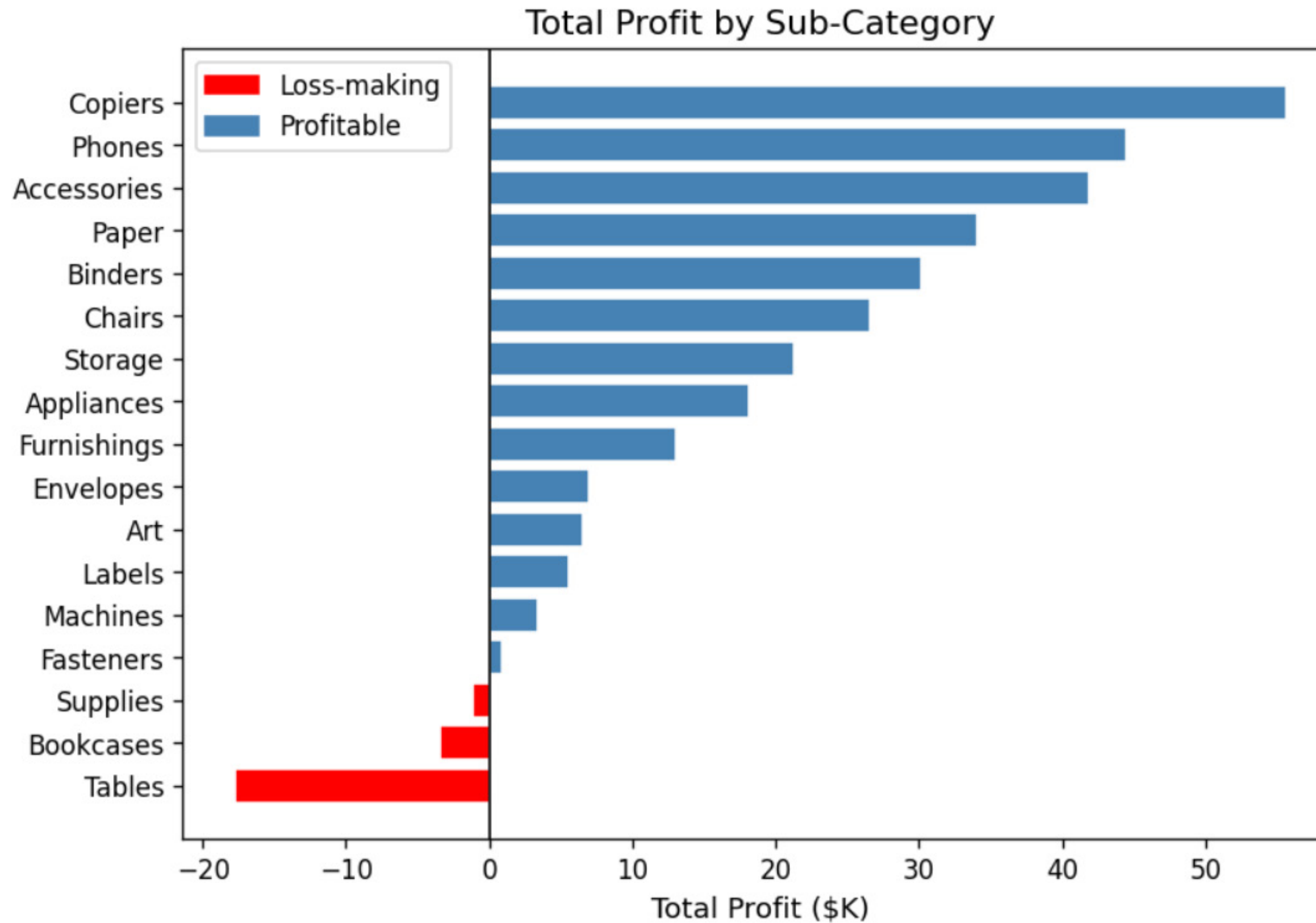
Phones and **Accessories:** strong positive

Tables: biggest loss at –\$18K

Bookcases: –\$3.5K structural loss

Supplies: –\$1.2K

Portfolio is deeply unbalanced



Sales and Profit by Region

INSIGHT

All four regions are profitable, but Central clearly underperforms. Its profit is disproportionately low relative to the sales it generates.

West: highest profit (\$108K) and best margin

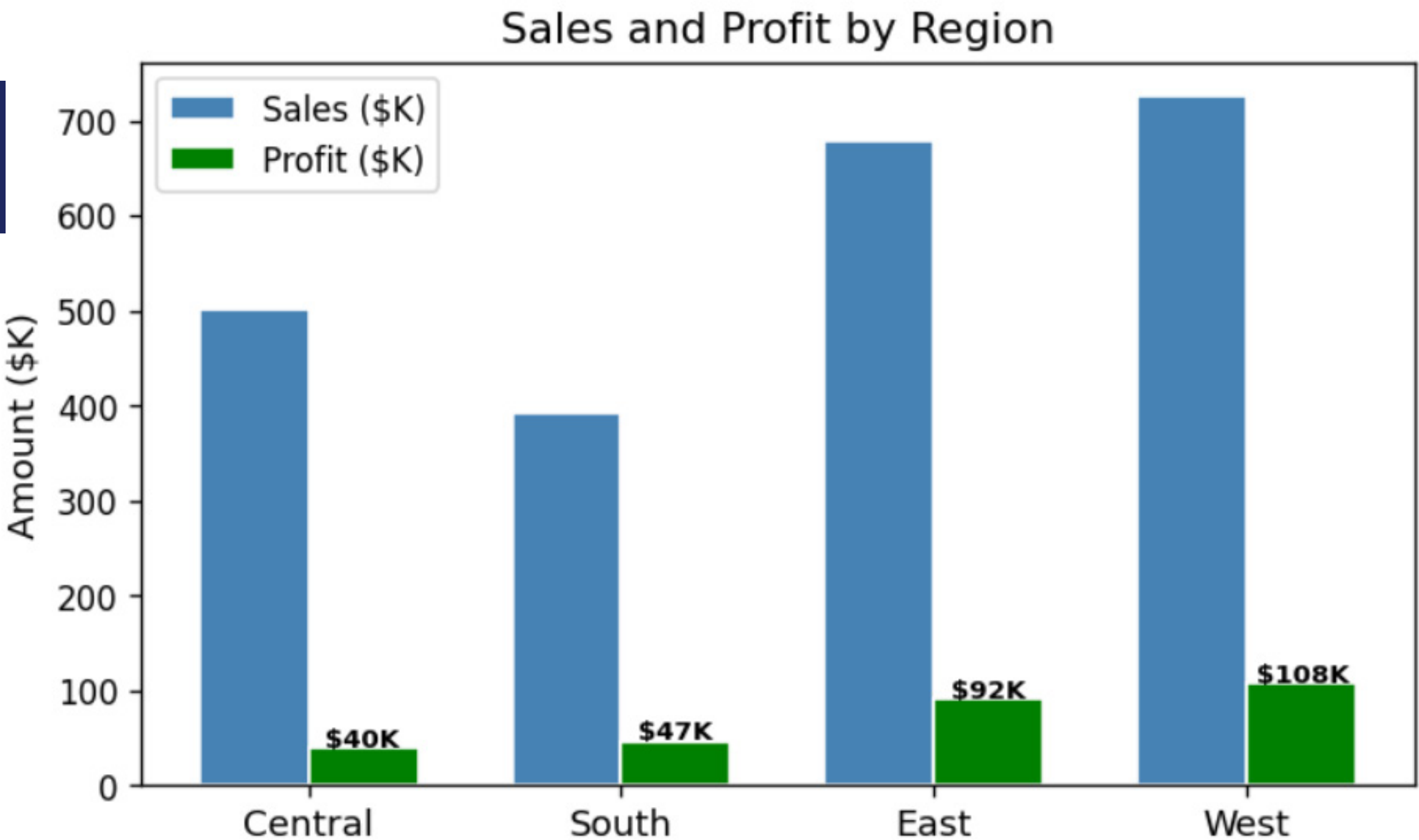
East: strong sales and solid profit (\$92K)

South: balanced performer (\$47K)

Central: lowest margin despite decent sales (\$40K)

East has high volume but margin below West — growth potential

Regional discount policy is likely inconsistent



Profit and Margin by Customer Segment

INSIGHT

All three segments are profitable, but Home Office punches above its weight on margin. It has the fewest customers yet the highest margin percentage.

Consumer: largest segment, most absolute profit

Corporate: solid middle ground

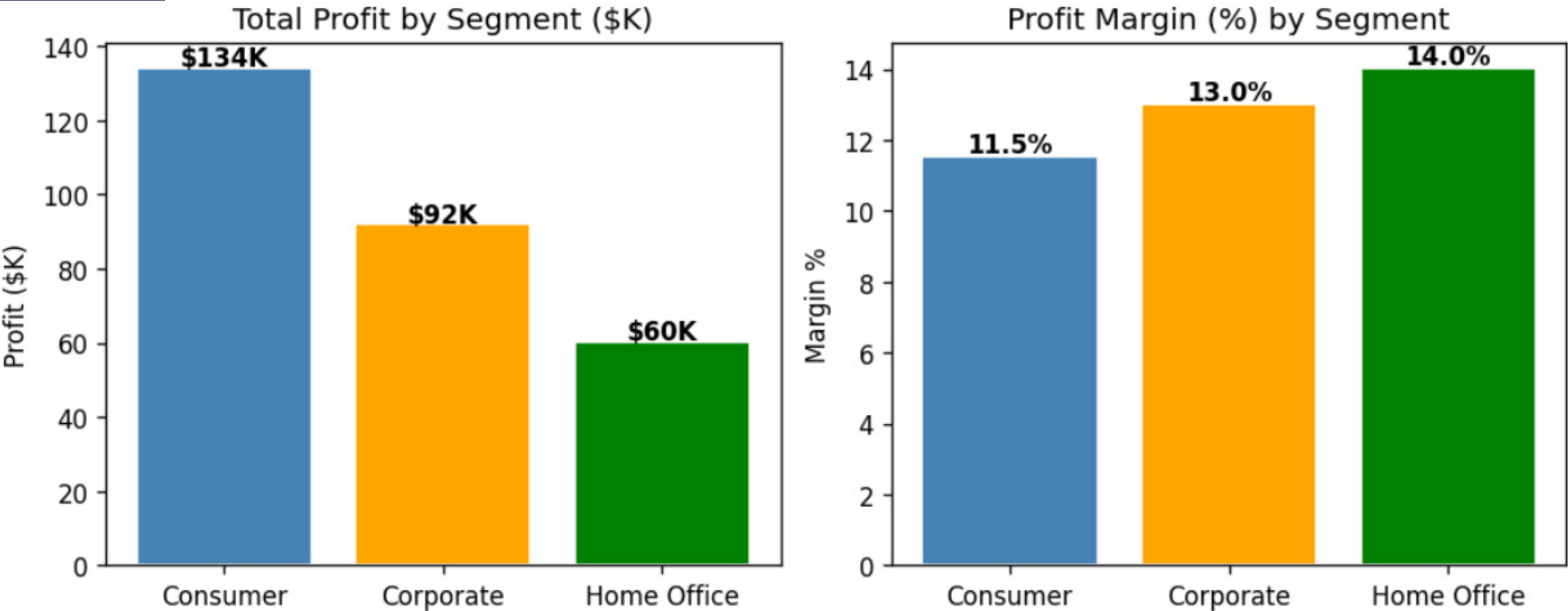
Home Office: 14% margin — best of all three

Home Office has only **148 customers**

Each Home Office customer is worth more in margin terms

Expanding this segment is a high-ROI opportunity

Segment Performance



Customer Profitability Breakdown

INSIGHT

1 in 5 customers is actively losing the business money. These customers are not marginal — they accumulate significant losses that cancel out profitable customer gains.

638 customers are profitable (80.5%)

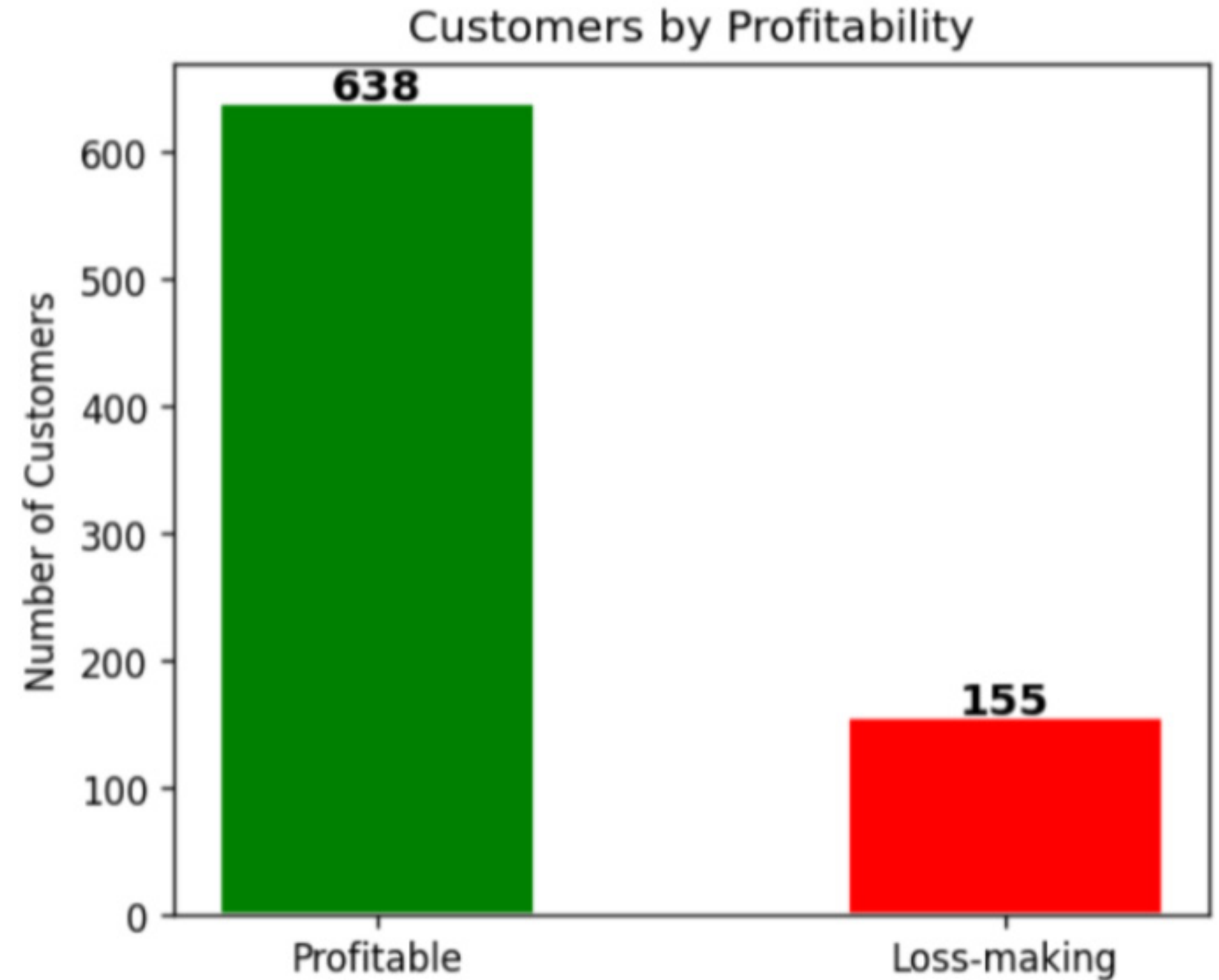
155 customers are loss-making (19.5%)

Total losses from loss-making customers: **-\$71,224**

Loss customers receive significantly higher avg. discounts

Removing discounts from loss customers alone = \$71K recovered

This is actionable without losing a single customer



Conclusions

- a) The 20% discount threshold is a hard boundary in the business. Below 20%, all segments are profitable. Above 20%, 9 out of 10 lines lose money.
- b) The concentration of damage is extreme. This is extraordinarily positive from an operational standpoint: the problems are localized, identifiable, and therefore can be addressed with surgical precision. There is no need to reinvent the company.
- c) The company has been looking at the wrong metrics: counting orders when it should be measuring margin by customer, by product, and by region.
- d) Discounts are not a commercial incentive, they are a structural leakage.



"Cindy Stewart, professional discount hunter. She never pays full price, always finds the biggest deals... and somehow manages to turn every purchase into a loss for the company."